Flowed Clover Topological Charge: Normalization and Verification

PyQUDA Measurement

October 2026

1 Scope and conventions

This implementation measures only a hadron-independent gluon loop. It does not extract a nucleon form factor or perform a continuum extrapolation. Use directions $(0, 1, 2, 3) = (x, y, z, t)$ and $\epsilon_{0123} = +1$. Hermitian generators satisfy $\mathrm{Tr}(T^a T^b) = \delta^{ab}/2$. For links $U_\mu = \exp(iagA_\mu)$ and the sum $C_{\mu\nu}$ of four oriented plaquettes based at the same site, define

$$A_{\mu\nu} = \frac{C_{\mu\nu} - C_{\mu\nu}^\dagger}{8} - \frac{\mathbf{1}}{3}\,\mathrm{Tr}\,\frac{C_{\mu\nu} - C_{\mu\nu}^\dagger}{8}, \tag{1}$$

$$H_{\mu\nu} = -iA_{\mu\nu} = a^2 g G_{\mu\nu} + O(a^4). \tag{2}$$

Here A is anti-Hermitian, H is Hermitian, and both are traceless. In particular $H_{\nu\mu} = -H_{\mu\nu}$.

2 The three-trace formula

The 24 nonzero permutations in the epsilon contraction fall into three pairings. Reversal within either antisymmetric pair cancels the corresponding epsilon sign; exchanging the pairs leaves their trace unchanged by cyclicity. Each pairing therefore contributes eight equal terms, giving

$$q_{\mathrm{lat}} = \frac{\epsilon_{\mu\nu\rho\sigma}}{32\pi^2}\,\mathrm{ReTr}(H_{\mu\nu}H_{\rho\sigma}) \tag{3}$$

$$= \frac{1}{4\pi^2}\,\mathrm{ReTr}(H_{01}H_{23} - H_{02}H_{13} + H_{03}H_{12}). \tag{4}$$

No commutativity assumption is needed. Since $\mathrm{Tr}(HK)^* = \mathrm{Tr}(KH) = \mathrm{Tr}(HK)$, the local density is real. It is invariant under local color conjugation and changes sign under reversal of one spatial axis. For $H_{01} = H_{23} = \mathrm{diag}(1, -1, 0)/2$ and other independent planes zero, $q_{\mathrm{lat}} = 1/(8\pi^2)$, fixing a simple normalization test.

The links already contain the coupling:

$$q_{\mathrm{lat}} = a^4 q_{\mathrm{phys}} + \text{cutoff corrections}, \qquad q_{\mathrm{phys}} = \frac{g^2}{32\pi^2}\epsilon_{\mu\nu\rho\sigma}\,\mathrm{Tr}(G_{\mu\nu}G_{\rho\sigma}). \tag{5}$$

If G includes g, omit the explicit g^2 in the last expression. No additional flavor factor, fermion-loop sign or EMT factor of two belongs in the implementation. QUDA's anti-Hermitian lower-plane convention maps to this formula using $A = iH$; two factors give the required minus sign.

3 Fourier transform and source alignment

For $V_s = L_x L_y L_z$, origin $o = 0$ and integer momentum $\mathbf{n}$, store

$$L(\mathbf{n}, t_f, \tau) = \frac{1}{V_s}\sum_{\mathbf{x}} e^{2\pi i \sum_j n_j (x_j - o_j)/L_j} q_{\mathrm{lat}}(t_f, \mathbf{x}, \tau). \tag{6}$$

Reality implies $L(-\mathbf{n}) = L(\mathbf{n})^*$ and $Q(t_f) = V_s \sum_\tau L(0, t_f, \tau)$. For a source $s = (\mathbf{s}, t_0)$,

$$L_s(\mathbf{n}, t_f, \tau) = e^{-2\pi i \sum_j n_j (s_j - o_j)/L_j} L(\mathbf{n}, t_f, (t_0 + \tau) \bmod N_t). \tag{7}$$

The gluonic time shift has no antiperiodic fermion sign. The disconnected building block with a compatible polarized two-point contraction is

$$C^Q_{3,\Gamma} = V_s\big(\langle C_{2,\Gamma} L_s\rangle - \langle C_{2,\Gamma}\rangle\langle L_s\rangle\big). \tag{8}$$

Configuration/stream identity and source correlations must survive averaging. This is not yet a complete kinematic ratio.

4 Flowed observable and RGI limit

At fixed positive physical t_f, take the continuum limit before $t_f \to 0$:

$$F_Q^{\mathrm{RGI}}(Q^2) = \lim_{t_f \to 0} \lim_{a \to 0} F_Q^{\mathrm{GF}}(Q^2, a, t_f). \tag{9}$$

The small-flow expansion identifies the RGI density, including the singlet axial-divergence mixing structure [2, 3]. No separate axial measurement or standalone $\overline{\mathrm{MS}}$ conversion is performed here. Finite-flow results must not be labelled already RGI; running and mixing can produce non-polynomial, logarithmic corrections in the extrapolation. Global continuum charge conservation along flow does not imply independence of local nonforward matrix elements. At finite spacing even the clover total charge has cutoff effects. Require a physical window with $a \ll \sqrt{8t_f}$ and a radius small relative to insertion/source separations.

In the proposed nucleon convention [1],

$$M_N^{-1}\langle N(p')|q_{\mathrm{top}}|N(p)\rangle = \bar{u}(p') i\gamma_5 u(p) F_Q(Q^2), \quad d_Q = -F_Q(0). \tag{10}$$

The forward pseudoscalar bilinear vanishes: extrapolate from finite transfer. In general $Q^2 = |\mathbf{p}' - \mathbf{p}|^2 - (E' - E)^2$. The Wick-rotation sign must be derived together with gamma matrices, epsilon and the chosen polarized projector; the explicit Minkowski $i\gamma_5$ is not an extra i multiplying the real Euclidean density. A Breit-frame density is not an ordinary relativistic probability density. Existing overlap/AWI calculations [4] guide signal construction, not direct flowed-clover matching.

5 Verification and cost

CPU tests compare all 24 epsilon terms with the compact expression on noncommuting matrices, test gauge conjugation, reflection, Hermiticity, tracelessness, identity links, Fourier pairing, source alignment, scheduling, root ownership and HDF5 metadata. GPU tests check actual link-level gauge invariance, MPI decompositions, QUDA total charge and the pre-extraction EMT baseline. Integration-step refinement compares equal physical output times; finite-step outputs need not be identical. See the companion Markdown validation summary for actual pass/pending status. The installed QUDA exponential has a singularity at exactly zero force. Identity links are therefore checked only at zero flow time; nontrivial fields exercise flow. This restriction does not exclude zero Fourier momentum or zero total charge, and does not constitute a fix of QUDA.

Six clover planes and three traces are needed per output time, with no quark inversions. Only projected loops are transferred and stored. An output with 125 momenta, 21 flow times and 80 time slices occupies 3.20 MiB in complex128; a full real 80^4 density would occupy 312.5 MiB per flow time.

5.1 Validated Aurora result

Job 8948692 completed successfully on 20 October 2026 with 10 nodes and 100 ranks. Its real 80^4 input was stream b, configuration 600, Coulomb gauge fixed with no additional HYP for topology. Both Wilson and Symanzik flow used 27 momenta and output times $t_f/a^2 = 0, 0.1, 0.2, 0.3, 0.4$. The pre-extraction EMT baseline was commit `5036ce0`; all compared EMT datasets agreed exactly. Small-field MPI-decomposition and gauge-transformation differences were below 1.92×10^{-16} and 1.01×10^{-15} respectively. Total-charge differences relative to native QUDA were at most 2.67×10^{-13} (Wilson) and 1.75×10^{-13} (Symanzik). Step halving from 0.01 to 0.005 to 0.0025 reduced consecutive loop differences by factors 8.22 and 8.36 respectively. The identity-flow limitation above remains in force; this is not a validation of that singular case.

Two clean repeats, excluding warmup and initial gauge input, took about 39.27 seconds per Wilson measurement and 41.02 seconds per Symanzik measurement. Separately instrumented passes assigned 32.56/34.28 seconds to flow and 6.56/6.56 seconds to clover construction. Density contraction was about 0.063 seconds, projection 0.016 seconds and HDF5 output 0.011 seconds or less. These are maximum accumulated component times over ranks, not strictly additive wall times. The companion Markdown report retains detailed results.

References

[1] Fukushima and Uji, *Polarized Nucleon as a Topological Dipole*, arXiv:2607.03123.

[2] M. Luescher, *On the chiral anomaly and the Yang–Mills gradient flow*, arXiv:2109.07965, Eq. (21).

[3] M. Luescher and P. Weisz, *Renormalization of the topological charge density in QCD with dimensional regularization*, arXiv:2103.15440, Sec. 6.

[4] M. Gong et al., *Strange and Charm Quark Spins from Anomalous Ward Identity*, arXiv:1511.03671.